Noise Reduction Using Cellular Automata

Naveen Chandra[1], Shashiraj Teotia[2], Ankur Chaudhary[3], Vikas Jain[4], Tarun Vashishtha[5], Nitya Nand Dwivedi[6]

[1,2,3,6] Department of Computer Application, Swami Vivekanand Subharti University, Meerut

[4] SCRIET, Ch. Charan Singh University, Campus Meerut, [5]IAMR Group of Institutions

naveen.svsu@gmail.com, shashirajt@gmail.com, ankurchaudhary849@gmail.com, ursjain@gmail.com, tarunvashishth@gmail.com, nityananddwivedi29@gmail.com

Abstract:

In this paper, we will eliminate the clamor present in a picture utilizing cell automata calculation. All are managing a Cellular Automata model worry with two primary clamors: one is salt pepper commotion and other is Gaussian commotion. For the most part the sound comes from when we post or snap a photo. Here on the off chance that we are discussing the decrease of commotion in computerized pictures, it is a piece of pre-handling sifting step of information. In ongoing year research, analysts have done grouping-based strategies in which they have first recognize the loud pixel and afterward supplant it with neighbor one yet they have not referenced any adjustment of the defiled pixels. The center channel can give a decent outcome in fundamentally the same as fields. The focal channel has a superior approach to obtain improved results than the focal channel in removing outside values in the sifting system without obscuring the picture.

Keywords: Cellular automata, Image Processing, Noise Filtering, PSNR.

DOI Number: 10.14704/nq.2022.20.6.NQ22456 **NeuroQuantology 2022; 20(6):4650-4672**

1. Introduction

Noise reduction is a signal removal process. All recording devices, both analogue and digital, have features that make it easy to make audio. The sound can be a random or white sound with no correlation, or a combined sound presented by the device machine or processing algorithms. In electronic recording devices, the most common type of noise is noise caused by random electrons, which are strongly influenced by heat, deviating from their fixed path. These missing electrons influence the output voltage of the output and thus create a visible noise. How-Lung Eng et al proposed a flexible novel modification modifier for soft-switching median (NASM) to effectively address the above-mentioned problems and achieve more advanced filtering efficiency in terms of efficiency in noise removal and strength against noise variability [1]. In the case of photographic film and magnetic tape, sound (both visual and audible) is introduced due to the grain structure of the medium and other hand Protocols play an important role in measuring network performance to control the data communication process [2]. In animated film, the size of the film's characters determines the sensitivity of the film, a highly sensitive film with large characters. In magnetic tape, when the particles of a magnet particle become larger, the medium becomes noisy. To compensate for this, large areas of film or magnetic tape may be used to reduce the noise to an acceptable level. There are two common types of sound: the same sound and the sound of salt & pepper. With the same sound, a distorted pixel can take any value from 0 to the maximum allowable value (we take an image with a gray scale). In the case of salt and pepper, a damaged pixel may take only one of two different colors: black or white. To remove unwanted sound and improve image quality, a central filter is used [2]. The central filter is a non-linear filter that is used to remove noise, the main disadvantage of which is that it obscures fine details or distorts the edges while filtering the sound. With a medium filter, the strength of the neighboring pixels is adjusted and the number of media is given to the average pixel.

Exploring the salt & pepper sound filter and Gaussian sound. Various sound values were added, and at each level the CA rules were applied 100 times. It has been found that using the new PSNR-conditioned CA method provided the best results, and unless otherwise stated all the results shown use this setting. Mobile automaton (CA) is a straightforward model studied in computability theory, mathematics, physics, complexity science, theory biology and microstructure modeling. Cellular automata are also called cell spaces, tessellation automata, equivalent structures, cellular structures, tessellation structures, and similar repetitive systems. A CA is a network of automata arranged in the same way, or cells, interacting with each other locally and with a specific shape.

Cellular Automata (CA) is a spatial computer model that provides an excellent platform for performing complex calculations with the help of local information. The CA is made up of interconnected cells, each containing an automaton, a simple machine capable of performing simple calculations [5]. Each automaton has a structure, which changes over time based on the regions of neighboring cells. The CA model modification law determines the neighborly relationship between automata. Each automaton changes its state (value) over time (t) based on the state in the previous period (t-1) of neighboring cells. Launched by John Conway in 1970, Game-of-Life is a well-known example of CA. CA has many applications for different fields. Because of the obvious similarities between 2-dimensional mobile automata and images (pixels insert a map into cells) the same mobile automata have been widely used in other areas for image processing or similar research. One of the reasons why cell automata has been chosen for this project is because very little work has been done in this area, which enables us to do real research rather than summarize and draw conclusions about someone else's work. In this project our aim was to demonstrate whether cellular automata are useful for image processing. The most important part of this project is to understand what Cellular Automata can and cannot do and why [7]. The actual piece of algorithm created is not so important, the fact that it actually works well is not important - it would be equally important if it did not work. In order to begin to determine whether a mobile automaton can be used for image processing we have had to do extensive research on mobile automata and evolutionary calculations, which have formed a large part of this project.

In computer science theory, automata theory is the study of mathematical objects called invisible machines or automata and computational problems that can be solved by using them. The word Automata comes from a Greek word meaning "self-made".
Figure 2.1 shows a state-of-the-art machine, which is one well-known part of automaton. This automaton consists of regions (represented by an image by circles), and a transformation (represented by arrows). As the automaton detects the input signal, it makes a change (or skip) to another position, depending on its function change (which takes the current position and the latest symbol as its input).

Because of the obvious similarities between automaton and images (map pixels in cells), automata has been widely used in other areas for image processing or similar research. A cell can be considered part of an automaton and its status changes depending on the environment. A 2-D image contains a 2-D series of pixels and these pixels are treated as cells to update their status on the basis of neighboring cells.

There are many cycles for this, yet all endeavors to decide if the real distinction in pixel values incorporate the sound or subtleties of the genuine pictures, as well as the underlying estimation while attempting to save the most recent [02]. The viability of the sound estimation calculation was tried utilizing both designated and free measures [03]. In computerized picture handling, each picture capability f (x, y) is characterized over a space rather than a persistent, and restricted or middle area [04]. As per its motivation, we contrast our model and standard channels as far as salt commotion and pepper [05]. A framework is typically a square tile, or decoration, of a few aspects; a few tailings do, however have not yet been utilized. Cell conditions not entirely settled by interfacing with adjoining cells. There are no accessible ways of discussing straightforwardly with far off cells [07]. The middle channel was once the most famous aberrant channel to eliminate dynamic sound because of its superb sound result ability and PC effectiveness [08]. The outcomes, notwithstanding, show that our model is superior to the past models with regards to two aspects, in particular, MSE and HD [09]. For instance, in the event that a picture is harmed by 20% of drive clamor, 10% of pixels in the picture are harmed by direct tension and 10% of pixels by motivation driving forces [11]. Assuming the qualities in the picture histogram are dark or white, we reason that the picture contains the sound of salt and pepper, in any case it contains another sound [12].If a mobile automaton is a universal computing class it means that it can mimic the same computational power as a Turing machine - this means that a class 4 mobile automaton has the same aggregation capabilities as any computer [13]. Important meaning depends on the types of activities of interest. A necessary condition for the presence of essentials is that f must be included instead of $[0, \infty)$ [15]. Intermediate filtering is widely used in digital image processing because, under certain conditions, it retains edges while removing audio (but see discussion below) [16]. The cell and its two neighbors form the nucleus of 3 cells, so there are 23 = 8 possible patterns in space [18]. Our CA model successfully removed two types of noise at different scales [19]. With regard to one possible extension of color image processing: color level images were not attempted because gray image processing would have to be terminated in order to use color image processing [20]. This comparison shows that a CA-based filter provides significant improvements in standard filtering methods [21].

2. Lecture Review

2.1. Noise in An Image

Picture Sound is unstructured, it isn't something in the image, light variety or variety data in pictures and it is generally a component of electronic sound. It very well may be delivered with the sensor and turn of a scanner or computerized camera. The picture can likewise be seen on film screen and the inescapable shooting sound of a decent photon finder. Picture sound is a disliked visual item that adds bogus and outer data.

The first importance of "sound" and stays "superfluous sign" for example undesirable electrical variances in radio-created signals cause discernible commotion or dry sound. By relationship the undesirable electrical vacillations came to be known as "commotion". The picture, obviously, is unintelligible [21]. The extent of the picture can go from practically undetectable dabs in computerized picture catch, to for the most part enlightened brilliant and radio pictures, from which a limited quantity of data can be gotten through complex handling (sound level wouldn't be OK by any means as it wouldn't be imaginable to figure out what the subject was).It is by and large positive that the radiance of the picture (or film thickness) be something very similar with the exception of where it changes to make the picture. There are highlights, in any case, that frequently produce contrasts in the splendor of the showed picture or no picture subtleties

accessible. These varieties are typically irregular and don't have a particular example. By and large, it diminishes picture quality and is particularly significant when the articles are little and have low splendor. This irregular variety in picture brilliance is a decent solid [2]. All clinical pictures contain visual keenness. The presence of sound gives the picture a brilliant, grain, sewn, or frozen look. Figure 3.1 beneath contrasts two pictures and different sound levels. The picture comes from various sources, as we will before long find out.There is no sound recording method, but sound is more common in some types of image processes than others.
Example-

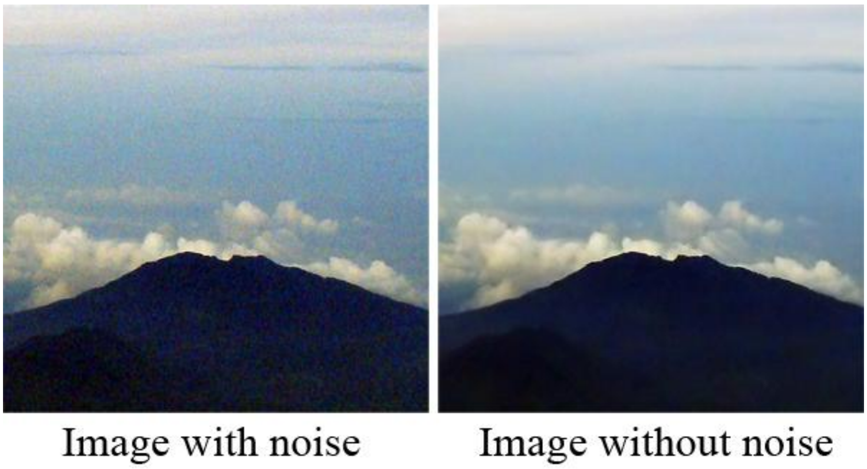

Fig2.1- Image with and without Noise

Nuclear images are often very noisy. Sound is also important in MRI, CT, and ultrasound imaging. Compared to this, radiography produces low-resolution images. Fluoroscopic images are slightly louder than radiographic images. Normal photography produces soundless images compared to where the film character is visible [3]. Although sound gives the image an unwanted appearance often, the most important thing is that the sound can block and reduce the appearance of certain features within the image. Visual loss is very important in things with low contrast. The visual limit, especially on objects with low contrast, depends largely on the sound. In fact, when we reduce the noise level of the image, the "curtain" is somewhat elevated, and many objects with low contrast within the body are visible. If the volume cannot be adjusted for a specific recording process, then why not reduce it to a minimum? While it is true that we can often change the characteristics of the images to reduce the noise, we must always be flexible. In x-ray imagery, the main concession is patient exposure and dose; in MRI and nuclear imaging, the main compromise has time to think [4]. There is also a correlation between sound and other elements of the image, such as brightness and blur. In fact, the user of each recording method must determine the acceptable sound quality of a particular process and select the image features that will achieve it with minimal exposure, image capture time or effect to other image quality features.

2.2. Sources of Noise in Digital Images

Digital images are prone to a variety of audio formats. Audio is the result of errors in the image capture process that result in pixel values that do not reflect the true thickness of the actual scene. There are several ways to present sound in an image, depending on how the image is made. For example:

• When a picture is scanned in a film, the character of the film is the source of the sound. The sound can also be the result of a film damage, or presented by the photographer himself.

• If the image is received directly in digital format, the data collection method may produce audio.

• Electronic image transfer of data can produce audio.

2.3. Categories of Noise

An image can be degraded by the following two ways:

1. Additive Noise
2. Multiplicative Noise

2.3.1. Additive Noise

This additional sound comes from a variety of sources, one of the main features being Temperature. The sound said more in the sense that this sound only adds to the signal not including the sound, so it is an integrated process that does not include one.

$$\mathbf{G_{ij}= F_{ij}+N_{ij}}$$

Where F_{ij} is the intensity of pixel (i, j) in the original image,

G_{ij} is the intensity of pixel (i, j) in the degraded image and

N_{ij} is the additive noise.

2.3.2. Multiplicative Noise

In signal processing, the term repetitive sound refers to a random unwanted recurring signal into a specific signal suitable during recording, transmission, or other processing. An important example is the sound of dots that are often seen on a radar image. Examples of duplicate sound affecting digital images are appropriate shades due to the flexibility on the surface of the imagery, shadow created by complex objects such as Venetian leaves and blinds, dark spots caused by dust in the lens or image sensor, and variations in individual parts of the image sensor. Repeated sound can be expressed as-

$$\mathbf{M_{ij}=U_{ij}*N_{ij}}$$

Where, U_{ij} is the intensity of pixel (i, j) in the original image,

M_{ij} is the intensity of pixel (i, j) in the degraded image and

Nij is the multiplicative noise.

Repetitive sound, also known as speckle sound, is dependent on signals and is difficult to remove. Based on the fourth version model, a new method of removing repetitive sound from images was proposed. In fact, the Fourier transform and logarithm technique is used in a sound image to convert the sound of convolution into an additional sound, so that the sound is removed using the traditional algorithm for removing additional noise from the frequency range.

eISSN 1303-5150

To remove noise, the improved model avoids the blocky effects produced by the second order model and acquires a better edge-saving ability. The performance of the proposed method is tested in images with additional sound and repetition. Compared with some traditional methods, the test results show that the proposed method achieves high performance at different PSNR values and viewing quality.

2.4. Types of Noises

Photo sound is a random variation of light or color information on images generated by the sensor and rotation of a scanner or digital camera. The sound of the image can also be seen on film screen and the inevitable shooting sound of a good photon detector. Image sound is often regarded as an undesirable product of photography. Although this unwanted variation is even known as “noise” in comparison to unwanted noise but it is not audible.

The types of Noise are following:-

1. Amplifier noise (Gaussian noise)
2. Salt-and-pepper noise
3. Shot noise(Poisson noise)
4. Speckle noise

2.4.1. Amplifier Noise (Gaussian Noise)

The most common model of amplifier sound is the add-on, Gaussian, independent of each pixel and independent of signal pressure, which is mainly caused by Johnson–Nyquist (hot noise), including that emanating from capacitor reset noise [21]. Sound amplifier is a major part of the "readable sound" of the image sensor, that is, the constant sound level in the dark areas of the image. Color-coded cameras are used to amplify the volume in a blue channel rather than in a green or red channel, it can make a lot of noise in a blue channel.

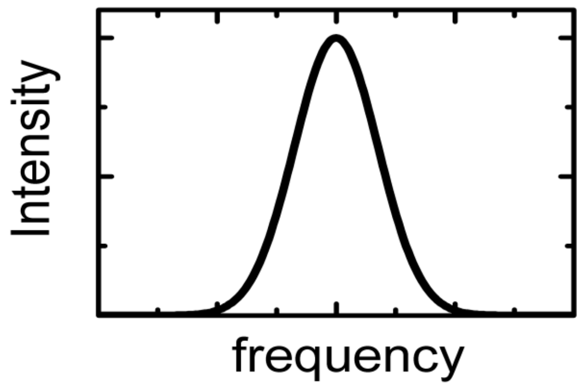

Fig2.2- Gaussian Noise Distribution

Gaussian sound is a mathematical sound that has the potential for compression function (pdf) for normal distribution (also known as Gaussian distribution) as shown in Figure 3.2 above. It is a large part of the "readable sound" of the image sensor, that is, the constant sound level in the dark areas of the image.

Effect of Gaussian Noise on an Image-

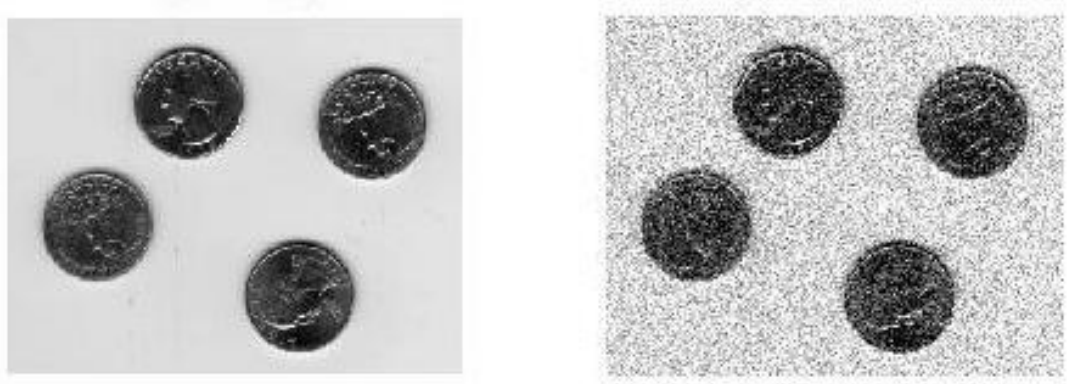

Fig2.3- Introducing Gaussian noise in an image

2.4.2. Salt-And-Pepper Noise

An oily or "irrational" tail is sometimes called the sound of salt and pepper or the sound of a spike. The vibrant image of salt and pepper will have dark pixels in bright areas and light pixels in dark areas. This type of noise can be caused by analog to digital translator errors, small transmission errors, etc. It can be effectively eliminated by removing the black frame and combining dark / light pixels.

Impulse noise is caused by pixel malfunction of the camera sensors, hardware memory errors, or noise channel transmission [8]. The two most common types of stirring sounds are the salt and pepper as well as random sounds. In images damaged by the salt-pepper sound (respectively, random value), noisy pixels can only take high values and small values (respectively, any random value) in a variable range. There are many tasks to restore images that have been damaged by intense noise. However, when the audio level is above 50%, some details and edges of the actual image are smeared with a filter.

The vibrant image of salt and pepper will have dark pixels in bright areas and light pixels in dark areas. This type of noise can be caused by dead pixels, analog-to-digital converter errors, minor transmission errors, etc. This can be largely eliminated by removing the black frame and combining dark / light pixels. Salt & Pepper Noise is usually caused by pixel malfunction of the camera sensors or time errors in the digital process.

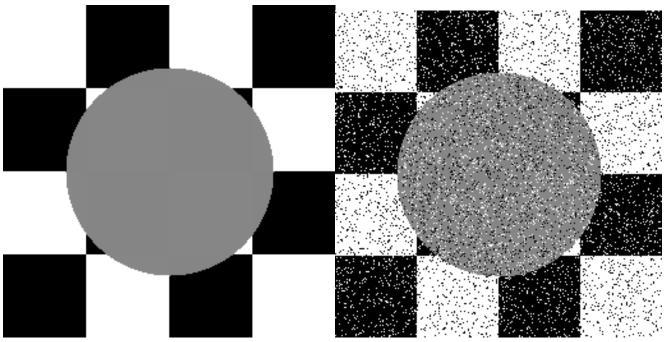

Fig 3.4- Effect of salt & pepper noise

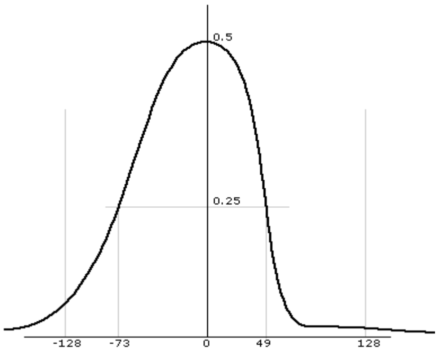

Fig 3.5- Salt & Pepper Noise distribution

2.4.3. Shot noise (Poisson noise)

Unnecessary commotion in straightforward pieces of the picture from the sensor of the picture is normally brought about by measurable vacillations, i.e., varieties in the quantity of photons heard at a specific degree of openness. This sound is known as a photon shot. Shooting sound has a root-mean-square worth equivalent to the square foundation of the picture's force, and the sounds at various pixels are free of one another [03]. The shot follows the dissemination of Poisson, which is normally entirely vague from the Gaussian. Poisson clamor or shooting commotion is a kind of electronic clamor that happens when a predetermined number of energy-conveying particles, like electrons in an electric field or photons in a visual gadget, are sufficiently little to give huge factual adaptability in estimation.
Notwithstanding the photon picture sound, there might be extra shooting commotion from the ongoing that releases dark to the picture sensor; this sound is once in a while known as "dark weapon sound" or "dark firearm sound" [04]. The ongoing haziness is excessively perfect for "hot pixels" inside the picture sensor. Adaptable dark charging of customary and hot pixels can be taken out (utilizing the "dark edge"), leaving just a strained sound, or an irregular part, of spillage. In the event that the evacuation of the dark edge isn't finished, or on the other hand assuming the openness time is long enough for the hot pixel charge to surpass the line charging limit, the commotion will be higher than the shooting clamor, and the hot pixels will show up as salt and pepper commotion.

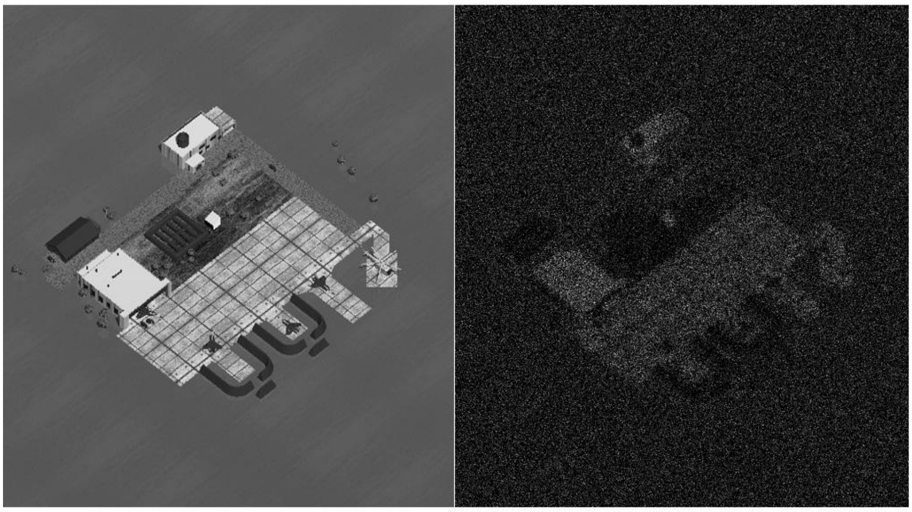

Fig 3.6- Effect of Poisson Noise on an image

2.4.4. Speckle Noise

Speckle sound is a naturally occurring granular and degrading active radar quality and synthetic aperture radar (SAR) images. The speckle sound on normal radar comes from random shift in the retrieval signal from something smaller than a single image processing device. Increases the average gray matter area. Speckle noise in SAR is often very sensitive, causing difficulty in interpreting the image. It is due to the parallel processing of back-to-back signals from multiple distributed targets [21]. In SAR oceanography, for example, speckle noise is caused by signals from primary scatters, gravity-capillary ripples, and is seen as a pedestal image, below the image of ocean waves.

Speckle sound in SAR is a repetitive sound, i.e. directly proportional to the gray area in any area. Signal and sound depend statistically on each other. Sample definition and single pixel variance are equal to the definition and variation of the area centered on that pixel.

Fig 3.7- Effect of Speckle Noise in an image

2.5. Estimation of Noise

Audio Scale for any image includes finding an image (or thumbnail) containing only sound, and then applying its histogram to the audio model. Sound effects can be obtained by directing the recording device (eg camera) to a blank wall. In the event that we are unable to detect audio images, a portion of the image is selected with a known histogram, and that information is used to determine the audio features. After the image component is selected, we extract the known values from the histogram, and the rest of our audio model [02]. In order to develop a valid model many small images need to be processed.

A measure of the spectrum of sound is usually found in the first few seconds of sound speech which are silent circuits. This assumption is valid in the case of dry sound where the sound spectrum does not vary significantly over time. The traditional detector also tracks audio only sound speech frames to update audio volume. But sound rating updates on those channels are limited to non-speech frames. This is not enough in the case of static noise where the sound spectrum of sound varies even during speech. So there is a need to update the audio spectrum continuously over time and this is done with an audio algorithm. This thesis proposes a new algorithm suitable for areas with constant noise. For target measurement, the error rate of the error-related error variant was calculated between the actual sound spectrum and the estimated white Gaussian sound, traffic noise and street noise with a 15 dB SNR using continuous English sentences and punctuation expressions.

2.6. Various Factors affecting Noise

There are various factors that affect the sounds present in an image. The items below determine the presence of sound in the image-

2.6.1. Pixel size

Basically, the bigger the pixel, the more photons it scopes, and for that reason it works on the sign to-clamor proportion (SNR) of a given openness. The quantity of electrons created by photons is corresponding to the tactile space and quantum effectiveness [21]. Sound power is additionally equivalent to the sensor region, yet electrical energy is equivalent to the square base of the power and region. At the point when you twofold the line size of a pixel, you twofold the SNR. The electron pixel limit is additionally equivalent to its area. This straightforwardly influences the powerful reach.

2.6.2. Sensor Technology And Manufacturing

A major technical problem is CMOS-related image filtering. Until 2000 CMOS was considered the worst noise, but it improved until the technology was comparable, differing only in detail. CMOS is less expensive because it is easier to add functionality to the sensor chip. Technology includes other aspects of sensory design and production, all of which will gradually improve over time.

2.6.3. ISO Speed

Advanced cameras control ISO speed by enhancing sign (and sound) in pixels. Consequently, the higher the speed of the ISO, the more awful the commotion. For full responsiveness the sensor ought to be tried at a couple of ISO speeds, including low and exceptionally high. The higher ISO had bigger, coarse saline grains, consequently delivering "strong" or grain [21] pictures. In computerized photography, a similar idea applies.when the ISO rating is low, the picture sensor turns out to be less delicate and thus has a smoother picture, since there is less computerized commotion in the picture. On the off chance that the ISO level is high (exceptionally delicate) the picture sensor should attempt to make a powerful picture, which delivers more computerized sound (those multi-hued specks in shadows and medium tones).

2.6.4. Exposure Time

Long openness with faint light will in general be more vocal than short openness to brilliant light, i.e., cognizance doesn't function admirably on sound. To completely mirror the sensor, it ought to be tried at long openness times.

2.6.5. Digital Processing

Sensors usually have 12-bit analog-to-digital (A-to-D) converters, so digital noise is usually not a problem at the sensor level. But when the image is converted to 8-bit (24-bit

eISSN 1303-5150

color) JPEG, the volume increases slightly [04]. The increase in noise can be very bad if a wide range of image manipulation is required. So it is best to convert to 16-bit files (48-bit color). But the slower depth of the output file makes little difference to the average (unadjusted) sound.

2.6.6. Raw Conversion

Green converters often use noise reduction (low pass filter) and sharpen, whether you want it or not; even if sharpening is off. This makes it difficult to measure the internal structures of the sensor.

3. Methodology

A digital image is considered a two-dimensional array of m * n pixels, each with a specific gray or color value. An image can be considered as a lattice setting of 2D CA where each cell corresponds to an image pixel, and in cases there may be different gray values or colors. For simplicity, an image of a 256-level gray scale in the Moore area (eight neighboring cells) is considered. The fixed value parameters are applied, i.e., the update rule applies only to non-boundary cells.
Moore neighbors are counted. All range values are sorted by ascending order. Then the smaller and higher values are removed in the form of filtered pixel values and the average pixel value is updated using CA law. The mask is moved to the next location and the review process continues until the final location of the audio image is reached. The revision of the average pixel value was made in terms of CA Act.
The results are compared to other available filtering methods according to the Peak Signal to Noise Ratio (PSNR).

3.1 Standard Filtering Techniques:

3.1.1 Mean Filter:

Medium filter (i.e.) smooth image data filter, thus removing noise. This filter performs a filter for each pixel image using gray level values in a square or rectangular window around each pixel [16]. The concept of filtering meaning is to replace each pixel value in an image with the value of its "mean" or "average" value, including its own. Specified filtering is generally considered as a convolution filter.

Unfiltered value

5	3	6
2	**1**	9
8	4	7

A

Mean filtered

5	3	6
2	**5**	9
8	4	7

B

Fig 3.1. Mean filter

3.1.2. Median Filter:

Central filter is an indirect digital filter, often used to remove noise. Such noise reduction is a common precautionary measure in order to improve the results of later processing (e.g., edge of the image).
The median filter is very effective in removing noise while not depleting sharp edges on the image. However, it usually does a better job than a straightforward filter for storing useful information in an image. The median "middle finger" is stronger than average.

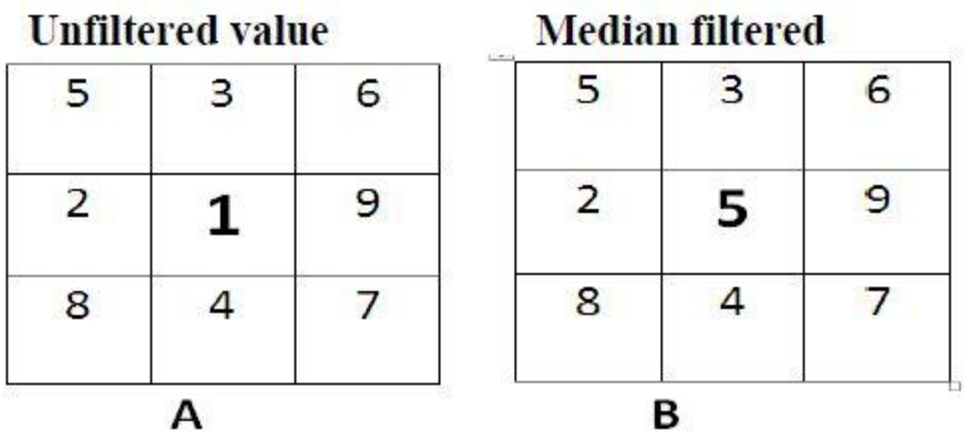

Fig 3.2. Median filter

2.2. Proposed New Cellular Automata Algorithm:

Introducing the CA noise model for removal novel. Our experimental model is mainly based on salt noise and pepper and Gaussian sound. First we introduce the type of noise explicitly through the MATLAB functions. The next step is to remove the noise using the CA rules defined in the given algorithm. Our planned CA model evaluates the sort of sound and reaction suitably for each kind. Presently, we do exclude the greatest and least qualities from the neighbors and ascertain the meaning of remaining qualities, after which we give the ongoing status of the robot.

Algorithm:

1) 2-D image m*n input
2) Pre-allocate another matrix with size (m+2,n+2) with zeroes.
3) Copy the input matrix in pre-allocated matrix.
4) Form a window matrix (3*3) of elements with the elements of input matrix.
5) Copy the window matrix into an array & sort it.
6) Eliminate the lowest & biggest element.
7) Find the average of all the remaining elements.
8) Replace the element with the average calculated.

9) Do the process for complete input matrix.
10) Change the image into color range type.
11) Display the noise reduced image.

3.3. Flow Chart:

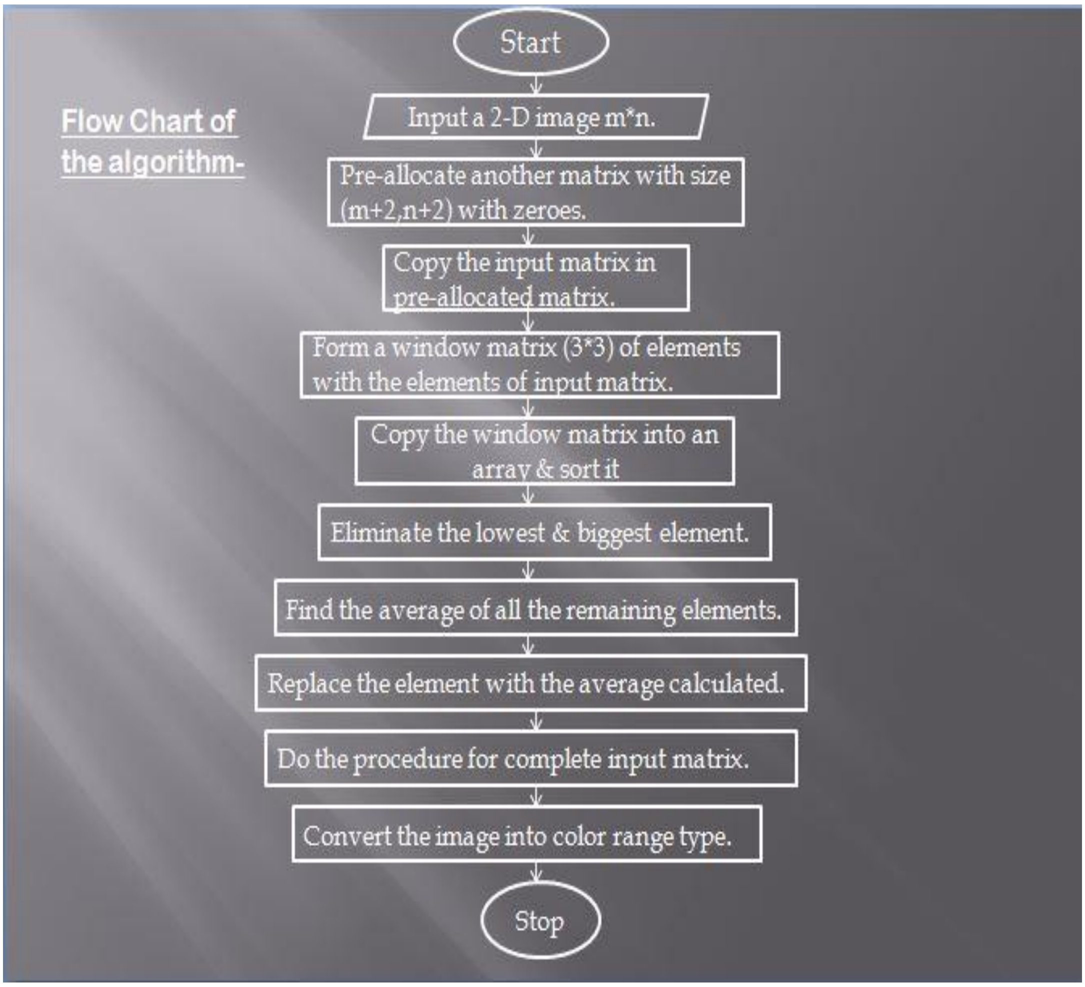

Fig 3.3. Flow Chart of the Proposed Algorithm

4. Results / Applications

We used the CA templates for our proposed vision, using the well-known software MATLAB-7.11.0 (R2010b Matlab, web index). We used one standard image in our test, which is a coin, and one of our pictures (Camera), which has more details and clearer images, and that is why we set a good example for testing. We utilized the very estimations that were utilized, in particular Mean Squared Error (MSE) and Hamming Distance (HD). It is known that when these means are little, the cycle is viewed as the best.
It ought to be noticed that MSE is significantly more precise than HD, in light of the fact that it computes the level of contrast between the two pictures, while HD just gives the quantity of various pixels in the two pictures. In each image I have added various rates of sound with aspects of 5%, 10%, 20%, 30%, 40%, and half of the picture size.
For each action we created two sound pictures: one with a sound "salt and pepper", and one more with a "Gaussian" sound. We utilized each picture reenactment multiple times and recorded the outcomes (fixed picture with two mistake estimations, MSE and HD). The

perplexing time is steady, O (1), because of the similitude given by the CA which is viewed as the primary benefit of the CA during activity. We ought to likewise take note of that the best estimation is the natural eye, particularly assuming that there is a reasonable distinction in wagering. At this stage, CA sound separating plans are assessed and contrasted and standard sifting methods in light of the greatest sign to-commotion proportion (PSNR). Sound scale is utilized to address how seriously the picture is harmed. The impacts of the depiction channel and the channel between the decreased sound are displayed in fig. 6.3 and. 6.2 separately. Figure 6.4 shows a picture that limits the picture structure factor utilizing our proposed CA Algorithm plans. Table-I shows the most extreme sign to-commotion proportion (PSNR) for the "Coins" picture.
It is clear from the results presented in Table 1 that the introduction of the CA Algorithm is better than the connecting CA Von Neumann channel. CA Algorithm gives further developed results than existing channels. Since in this structure nearly nothing and high pixel values are eliminated and the typical pixel regard is changed using CA guideline.

4.1 Computation for Salt & Pepper Noise:

Following Table shows the comparative study of PSNR values of Obtained images after the filtering technique.

Noise Intensity	Median Filter	Average Filter	CA Algorithm
1%	24.6706	24.5333	24.8010
5%	18.0319	18.4531	18.3650
10%	15.2292	15.5712	15.4021
20%	12.1994	12.7866	12.6030
30%	10.4086	11.0161	10.9886
40%	9.2036	9.8859	9.8622
50%	8.1867	9.0248	8.9728

Table I- Comparative Study of salt & pepper noise with various Filters

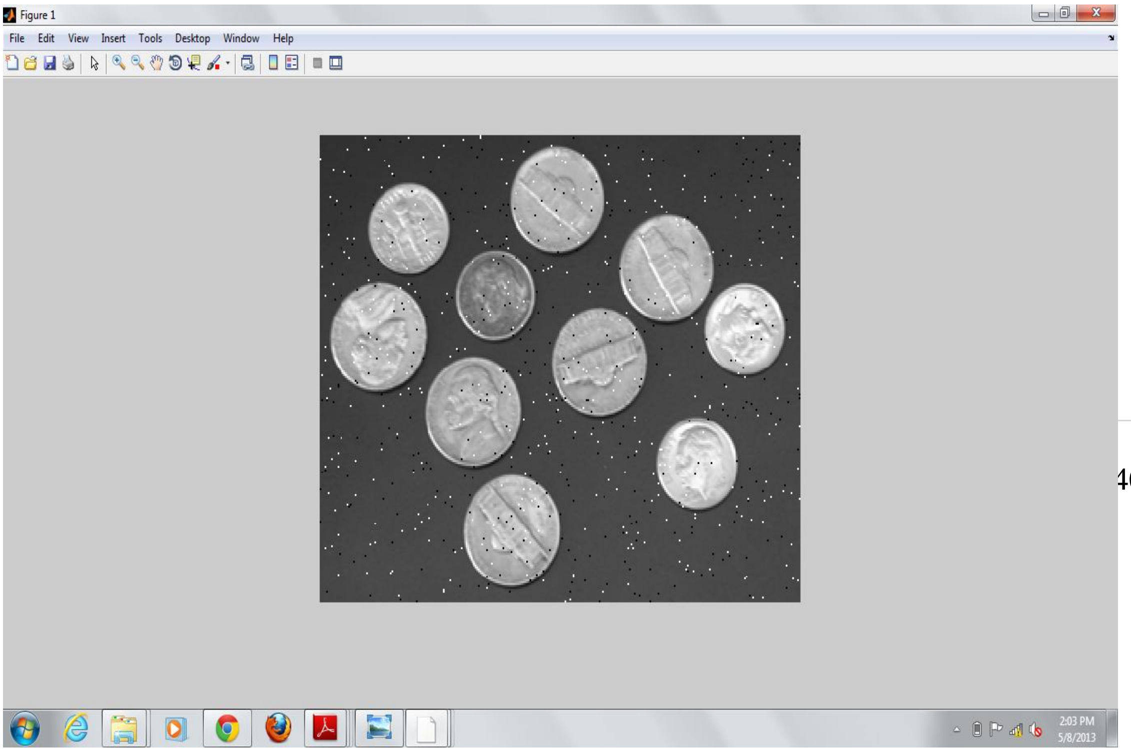

Fig.4.1 Image distorted by Salt & pepper Noise

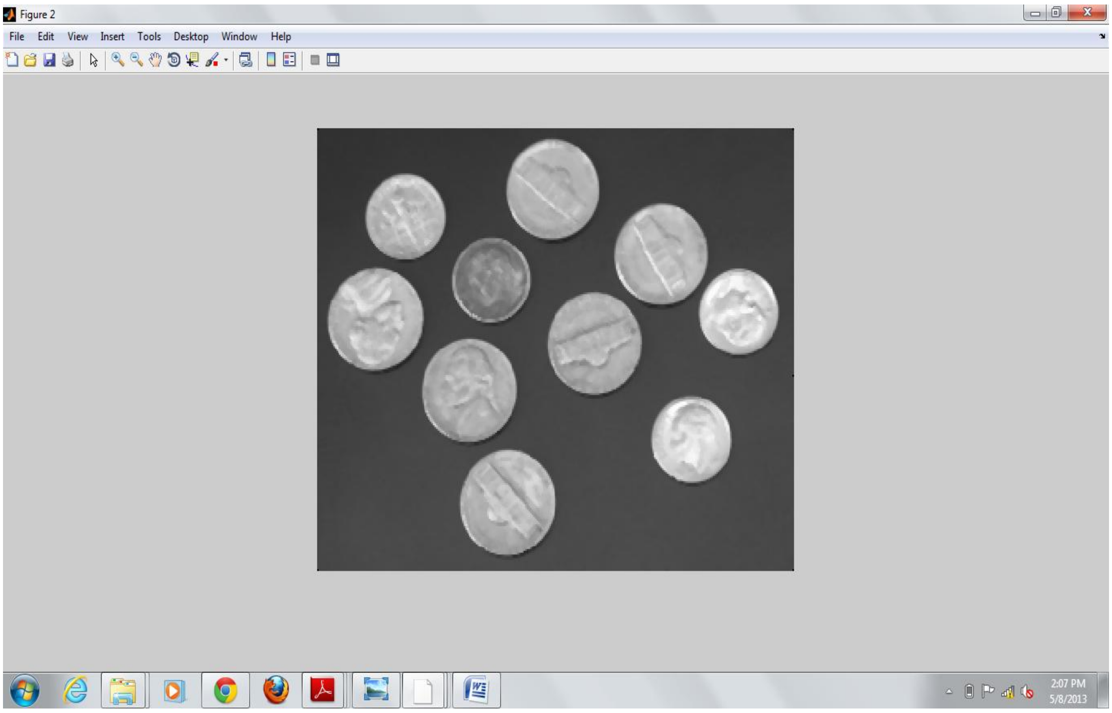

Fig.4.2 Image Filtered by Median Filter

Fig.4.3 Image Filtered by Average Filter

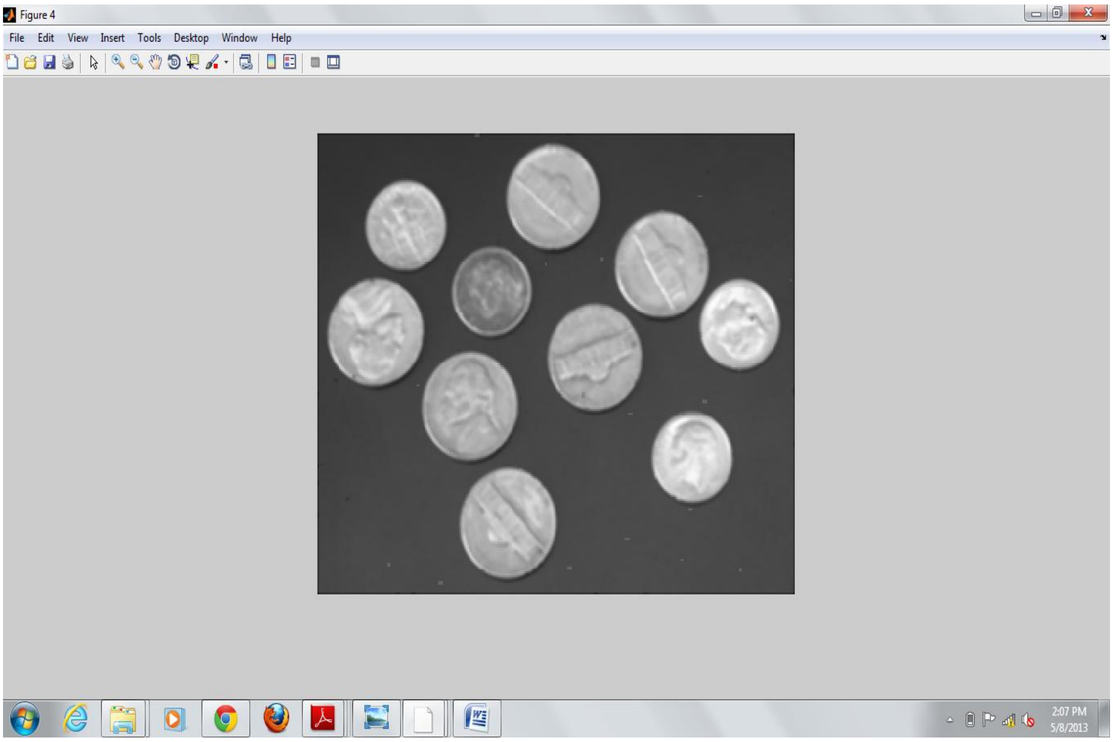

Fig.4.4 Image Filtered with CA based Algorithm

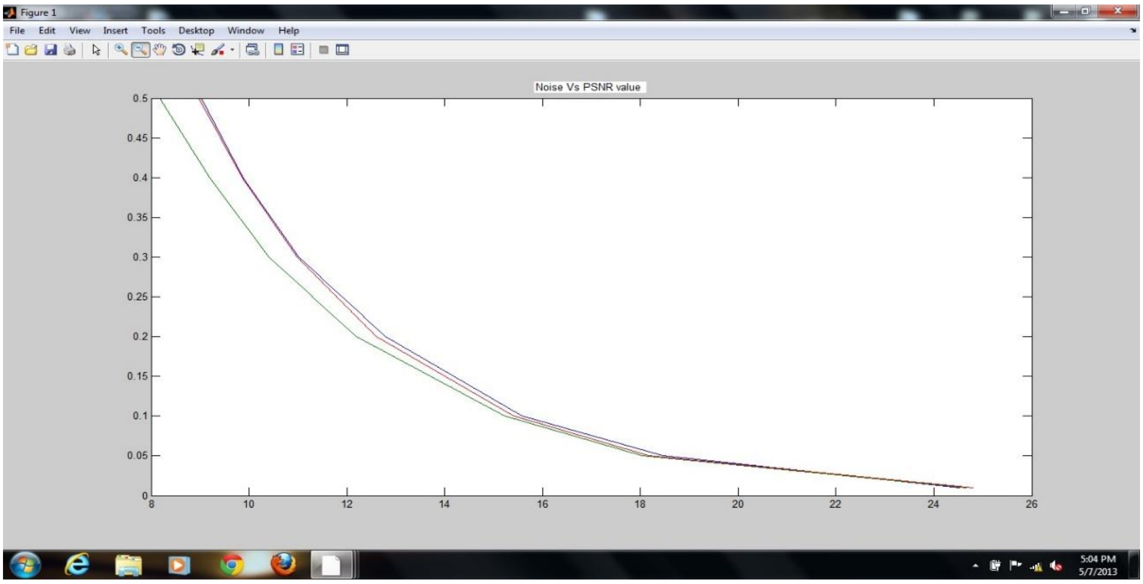

Fig.4.5 Salt & pepper Noise Vs. PSNR values

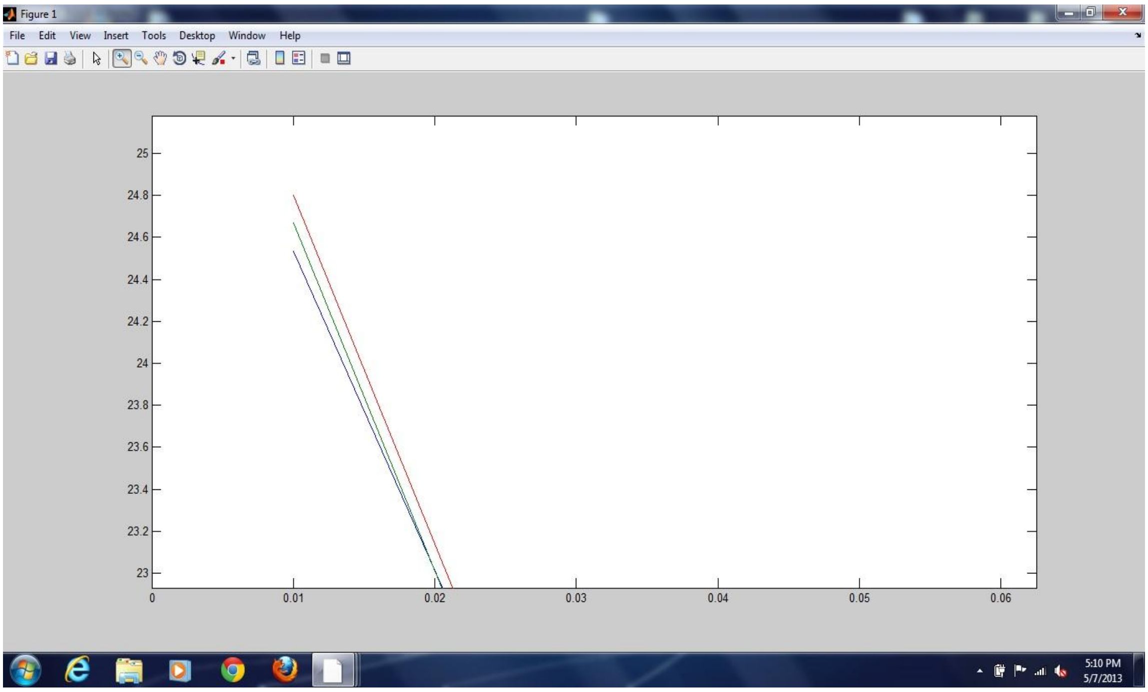

Fig.4.6 PSNR value Vs. S&P Noise

(Red-CA based Filter, Green-Median Filter, Blue-Average Filter)

4.2 Computation for Gaussian Noise

Following Table shows the comparative study of PSNR values of Obtained images after the filtering technique.

Noise Intensity	Median Filter	Average Filter	CA Based Filter
1%	20.1311	20.2281	20.2298
5%	20.1800	20.2048	20.2900
10%	20.2678	20.3207	20.3117
20%	20.5867	20.5273	20.5510
30%	21.0633	20.9490	21.0273
50%	21.5170	21.3868	21.4700
60%	21.7604	21.3819	21.5294

Table II- Comparative study of Gaussian noise with various Filters.

Fig.4.7 Image distorted by Gaussian Noise

Fig.4.8 Image Filtered by Median Filter

Fig.4.9 Image Filtered by Average Filter

Fig.4.10 Image Filtered with CA based Algorithm

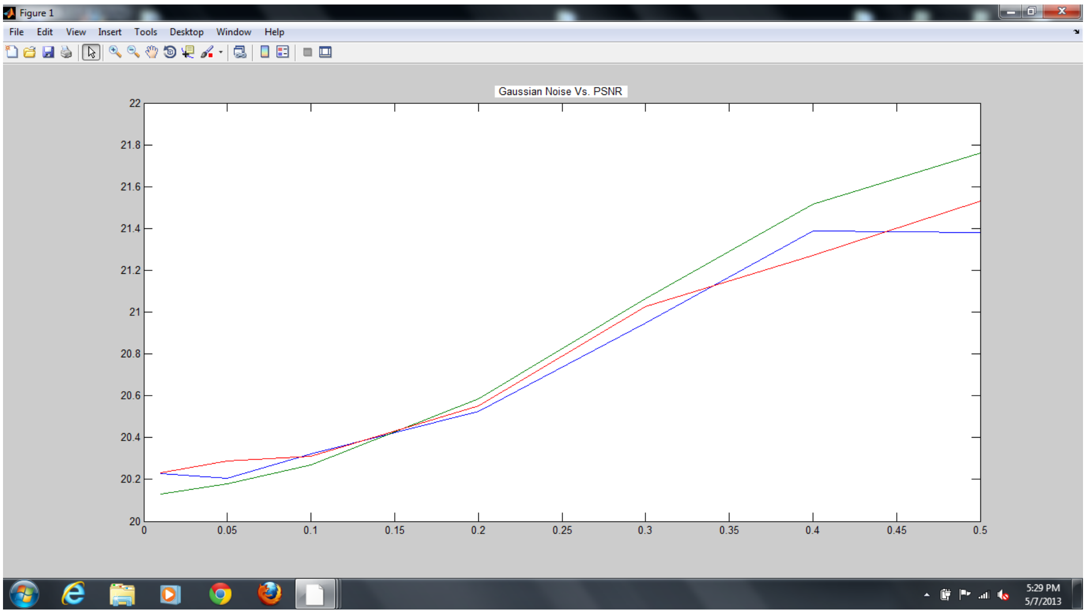

Fig.4.11 PSNR value Vs. Gaussian Noise Intensity

(Red-CA based Filter, Green-Median Filter, Blue-Average Filter

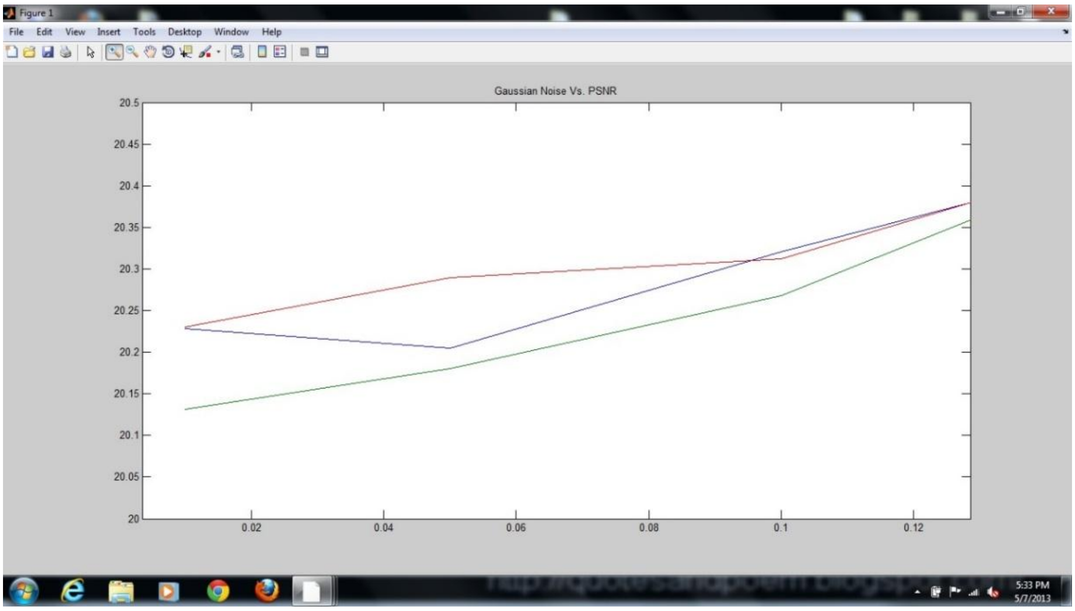

Fig.4.12PSNR value Vs. Gaussian Noise

(Red-CA based Filter, Green-Median Filter, Blue-Average Filter)

5. Conclusions

In this paper we have presented a CA model for picture sound expulsion. Our model works effectively with both the "salt and pepper" sound and the "gaussian" sound. We have shown that our model is superior to the past model, and is nearly on par with the Median channel.
As a future project, we will first develop the proposed current model in terms of performance and accuracy, without making it general to deal with additional audio types. With regard to the initial phase of clarification, mainly gray images were used in mobile automata to perform image processing. Image color in specification refers to RGB color images. Second, this study largely leads into the future the use of two-dimensional mobile automata in other applications without image processing, as image processing is not really a difficult computer task to utilize the full “power” of unbalanced cell phones. automata therefore the following is the framework in which the study provided some two-dimensional ideas of mobile automata. Some use of mobile automotive photography creates “live” effects. Image as an example, instead of a chrome logo or a light effect of cell automata can produce a smooth curved area made up of small moving “mosaic”.

6. References:

1. Eng H. and Ma K., 2000, "Noise adaptive soft-switching median filter for image de-noising", *IEEE International Conference on Acoustics, Speech, and Signal Processing*, Vol. 4, 2000, pp. 2175-2178.

2. Nitya Nand Dwivedi and Zaidi, Taskeen, “Performance Analysis of Distributed Networks”, International Journal of Advanced Science and Technology, Vol. 29, No.3,pp.3283-3300,2020.

3. International Journal of Emerging Technology and Advanced Engineering Website: www.ijetae.com (ISSN 2250-2459, Volume 2, Issue 7, July 2012).

4. Cellular automata-based identification & removal of impulsive noise from corrupted image Journal of Global Research in Computer Science Volume 3, No. 4, April 2012.

5. Rafael C. Gonzalez and Richard E. Woods, "Digital Image Processing", 2001, pp.220 – 243.

6. D.R. Chowdhury, I.S. Gupta and P.P. Chaudhury, "A class of two-dimensional cellular automata and applications in random pattern testing", Journal of Electronic Testing: Theory and Applications 5, 65-80, (1994).

7. P. P. Chaudhuri, D. R. Chaudhuri, S. Nandi, S. Chatterrjee, "Additive Cellular Automata, Theory and Applications", Vol. 1, *IEEE Computer Society Press*, Los Alamitos, California, ISBN-0-8186-7717-1. (1997).

8. W. Pries, "A Thanailakis and H.C. Card, Group properties of cellular automata and VLSI Application," *IEEE Trans on computers C-35*, 1013-1024, Dec 1986.

9. Chih-Yu Hsu, Ta-Shan Tsui, Shyr-Shen Yu, and Kuo-Kun Tseng," Salt and Pepper Noise Reduction by Cellular Automata",*International Journal of Applied Science and Engineering*2011. 9, 3: 143-160

10. Rosin, P. L. 2006. "Training Cellular Automata for Image Processing", *IEEETransactions on Image Processing*, 15: 2076-2087.

11. Rosin, P. L. 2010. "Image Processing using 3-State Cellular Automata", *ComputerVision and Image Understanding*, 114: 790-802.

12. P. P. Choudhury, B. K. Nayak, S. Sahoo, S.P. Rath, 2008. "Theory and Applications of Two-dimensional, Null-boundary, Nine-Neighborhood, Cellular Automata Linear Rules", in: arXiv: 0804.2346, cs.DM; cs.CC; cs.CV. (2008).

13. New Image Noise Reduction Schemes Based on Cellular Automata,*International Journal of Soft Computing and Engineering (IJSCE)* ISSN: 2231-2307, Volume-2, Issue-2, May 2012.

14. D.R. Chowdhury, I.S. Gupta and P.P. Chaudhury, "A class of two-dimensional cellular automata and applications in random pattern testing", *Journal of Electronic Testing: Theory and Applications* 5, 65-80, (1994).

15. Abdul Raouf Khan, "On Two Dimensional Cellular Automata and its VLSI applications", Department of Computer Sciences, KING FAISAL UNIVERSITY, SAUDI ARABIA.

16. T. Sunand Y. Neuvo, "Detail-preserving median based filters in image processing," *Pattern Recognition Letters*, vol.15, no. 4,pp. 341–347, 1994.

17. H. Hwangand, R. A. Haddad, "Adaptive median filters: new algorithms and results," *IEEE Transactions on Image Processing*, vol.4, no.4, pp.499–502, 1995.

18. Von Neumann, J.: "Theory of Self-Reproducing Automata", chapter in Essays on Cellular Automata. *University of Illinois Press, Urbana, Illinois*, 1970.

19. N.H. Packard and S. Wolfram, "Two dimensional cellular automata," *Journal of Statistical Physics*, 38 (5/6) 901-946, 1985.

20. N. Ganguly, P. Maji, S. Dhar, B. K. Sikdar, P. P. Chaudhuri, "Evolving Cellular Automata as Pattern Classifier", ACRI 2002, LN CS2493, Springer-Verlag Berlin Heidelberg (2002) pp. 56-68.

21. P. P. Chaudhuri, D. R. Chaudhuri, S. Nandi, S. Chattrrjee, "Additive Cellular Automata, Theory and Applications", Vol. 1, *IEEE Computer Society Press*, Los Alamitos, California, ISBN-0-8186-7717-1. (1997).

22. Peters, R.A., "A new algorithm for image noise reductionusing mathematical morphology", *IEEE Transaction onCommunication*. v. 4, n. 5, 554-568, 1995.